

Hyeonji Julia Lee

hyjulee@umich.edu | [LinkedIn](#) | [Personal Website](#)

EDUCATION

Master of Science in Information

University of Michigan, School of Information

Expected May 2028

Bachelor of Science in Computer Science

Washington State University

May 2026

RESEARCH INTERESTS

Human-Centered AI System Design

AI Literacy & Workforce Reskilling

Equity in the Future of Work

RESEARCH EXPERIENCE

Research Assistant

Aug 2026 – Present

Research Experience Development Program (REDP), School of Information, University of Michigan

- **Scaling Online Career Messages:** Supporting a qualitative study of career advice content on TikTok led by Ph.D. candidate Shanley Corvite, contributing to the AI-assisted analysis pipeline and usability evaluation with career practitioners.

Research Affiliate

May 2026 – Present

Center for AI Research in Education (CAIRE), Washington State University

- **Benchmark Research:** Led comprehensive data analysis and end-to-end manuscript writing as first author under the guidance of Dr. Li for a peer-reviewed paper on the ADAPT-AI benchmark.
- **ADAPT-AI & AI Group Coordinator:** Upgrading core features across both systems beyond their initial releases, including frontend UI/UX redesign and prompt atomization.

Research Assistant

Oct 2024 – May 2026

Center for AI Research in Education (CAIRE), Washington State University

- **ADAPT-AI (Founding Developer):** Built a Microsoft AI for Good-supported multi-agent assessment platform for rural K–12 science teachers. Designed and implemented the system architecture, full-stack API layer, and Azure integration. Currently in pilot testing with teachers.
- **AI Group Coordinator (Founding Developer):** Built a full-stack, multi-agent system, also supported by Microsoft AI for Good, that scaffolds real-time scientific argumentation in middle school classrooms. Designed to augment rather than replace teacher facilitation, improving transparency and fairness in classroom discussion.
- **Automated Assessment Research:** Engineered an OpenAI-based pipeline for automated concept map evaluation. Validated AI scoring reliability against human rubrics using image preprocessing and voting-based data aggregation.

CONFERENCE PAPERS AND PRESENTATIONS

- [3] Lee, H. J., Wang, Z., Chen, Z., Liu, H., He, P., & Li, T. (2026). *Toward Development: From Customizing Classroom Assessments with ADAPT-AI to Further Supporting 3D Learning Progression*. Paper presented at National Association of Research in Science Teaching (NARST). Seattle, WA.
- [2] Jin, G., Li, T., Lee, H. J., Xue, Y., He, P., Adesope, O. O., Dydasco, C. G., Sunday, O. J., & Nishida, K. (2026). *Using large language model to analyze chemistry undergraduate students' self-constructed concept maps*. Paper and poster presented at American Educational Research Association (AERA) Annual Meeting. Los Angeles, CA.
- [1] Lee, H. J., & Jung, Y. (2025). *Korean and Japanese Student Flows to the U.S. and International Research Collaboration Patterns*. Paper and oral presentation delivered at the 51st Japanese Modern Association of Korea (JMAK) International Conference. Osaka, Japan.

UNDER REVIEW

One first-authored paper proposal is currently under blind review for the American Educational Research Association (AERA) 2027 Annual Meeting.

WORKSHOPS

- [2] Li, T., Krajcik, J., Akgun, S., **Lee, H. J.**, & Wang, Z. (2025). *Designing 3D Assessments with Generative AI: A Hands-On Workshop for Elementary and Middle School Educators*. Full-day workshop delivered at National Science Teacher Association (NSTA) Professional Learning Institute (PLI). Minneapolis, MN.
- [1] He, P., Li, T., **Lee, H. J.**, & Wang, Z. (2025). *Designing 3D Assessments with Generative AI: A Hands-On Workshop for Elementary and Middle School Educators*. 4-hour workshop delivered at Washington Science Teacher Association (WSTA). Tacoma, WA.

TEACHING EXPERIENCE

Graduate Student Instructor Fall 2026
School of Information, University of Michigan

- **SI 504: Servers, The Shell, and Git.** Graduate-level course covering the Linux shell, remote server administration, and version control with Git.

FUNDING AND AWARDS

Graduate Student Instructor Appointment (0.50), University of Michigan Fall 2026
Full tuition waiver, stipend, and health benefits.

Boeing-Sponsored TMP Mentee Scholarship, Washington State University Fall 2024
Awarded through the Team Mentoring Program for engagement and professional development.

PROFESSIONAL EXPERIENCE

Archival Digitization Assistant Mar 2025 – May 2026
Manuscripts, Archives & Special Collections, Washington State University

- Digitized and cataloged historical newspaper clippings (1910s–1930s) for the Kimble Project, creating precise metadata to improve preservation and discoverability of regional archival records.
- Supported researchers at the MASC front desk, providing access to special collections and coordinating with subject librarians for advanced research requests.

Backend Developer May 2022 – June 2023
PCN, Seoul, Republic of Korea

- **Integrated Support Platform Development:** Engineered an industry information management system using Java and MySQL, leading project planning and end-to-end development to support post-COVID industry recovery.
- **Metaverse Management System Implementation:** Architected and restructured a system using PostgreSQL and open-source technologies, enabling administrators to create and manage virtual spaces for hybrid exhibitions.

SOFTWARE AND SYSTEMS

HUSS Performance Indicator Dashboard July 2026
Full-stack web application

- Built and deployed a system that tracks 19 performance indicators across five semesters for a five-university consortium, replacing a spreadsheet-based reporting workflow.
- Sole developer; in active use by administrators at all member institutions.

Contributor, TEAMMATES March – April 2026
Open-source peer feedback platform for education originating at the National University of Singapore, used by universities worldwide

- Contributed two pull requests merged into the main codebase (Java/TypeScript) after maintainer code review, working through the project's branch, pull request, and review workflow.

PROFESSIONAL SERVICE

Presenter & Interpreter, IFLA World Library and Information Congress (WLIC) 2026

Aug 2026

- Represent the Seoul Metropolitan Library, delivering the English-language poster session presentation.
- Provide English–Korean business interpreting for the library director across the three-day congress.

LEADERSHIP AND UNIVERSITY SERVICE

Board Member, Korean Student Association – Graduate (KSAG)

Aug 2026 - Present

- Plan monthly events for KSAG, a student organization founded in 1988 that serves roughly 500 Korean graduate students, postdoctoral fellows, and visiting researchers at the University of Michigan - over 1,000 people counting spouses and children.
- Design information sessions, networking events, and social gatherings that help members and their families settle into Ann Arbor.

Chair, Undergraduate Student Director, Students Book Corporation

Aug 2024 – Aug 2026

- Presides over board meetings to define semester goals and build partnerships with campus organizations (ASWSU, GPSA) to align services with student needs.
- Leads campus-wide outreach as PR Committee Chair, and manages recruitment and mentorship through the Search Committee to ensure leadership continuity.

Officer, Society of Women Engineers

Aug 2023 – Mar 2026

- **Large-Scale Event Coordination:** Led recruitment and management of STEM and architecture student clubs for KSED, overseeing safety certifications, venue logistics, and day-of operations.
- **Industry and Volunteer Leadership:** Managed professional outreach to recruit mentors and industry leaders, and led registration and operations committees for EWI and KSED initiatives.

Mentorship Program (TMP, Voiland Peer)

Aug 2023 – May 2025

- Served as both mentee and mentor in the Voiland Peer Network, providing tailored guidance to international, adult, and transfer students during their transition to the university.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, C#, TypeScript, JavaScript, SQL, HTML/CSS

Backend: FastAPI, Django REST Framework, Spring Boot, Node.js, SQLAlchemy, Pydantic

Frontend: React, Vite, Tailwind CSS, Radix UI, Shadcn/UI, React Hook Form

AI & ML: OpenAI, Multi-agent Systems, GenAI, PyTorch (CNN), NLP

Cloud & DevOps: Azure, Docker, AWS, Nginx, Git, GitHub

Database: PostgreSQL, MySQL, MariaDB, MongoDB

CERTIFICATES

- Google AI Professional Certificate (Google)
- Fundamentals of Deep Learning (NVIDIA Deep Learning Institute)
- Generative AI for Educators & Teachers (Vanderbilt University)
- NCS (National Competency Standards) based Training Course in Big Data Development Practice using Java & Python
- Microsoft Office Specialist Master Certification (MOS) 2016 (Microsoft)

REFERENCES

Tingting Li, Assistant Professor (Co-Director, CAIRE Research Lab)

College of Education, Sport, and Human Sciences, Washington State University
tingting.li1@wsu.edu

Peng He, Assistant Professor (Co-Director, CAIRE Research Lab)

College of Education, Sport, and Human Sciences, Washington State University
peng.he@wsu.edu

Parteek Kumar, Associate Professor

School of Electrical Engineering and Computer Science, Washington State University
parteek.kumar@wsu.edu

Joleen Warner, Digital Projects Manager

Libraries, Manuscripts Archives and Special Collections, Washington State University
joleen.stahl@wsu.edu